



Yakeen Plus (2025)

Practice Test - 08

DURATION : 200 Minutes

DATE : 13/11/2024

M. MARKS : 720

Topics Covered

Physics :	Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics, Alternating Current
Chemistry :	Practical Organic Chemistry, Coordination Compound, The d and f-Block Elements, Biomolecules
Biology :	(Botany): Principle of Inheritance and Variation Microbes in Human Welfare Molecular Basis of Inheritance (Zoology): Human Health and Diseases, Reproductive Health

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

SECTION - A

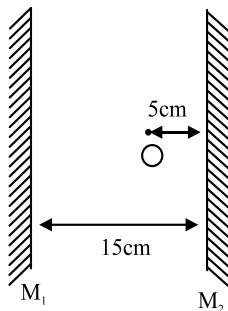
1. Two plane mirrors should be placed at what angle so that the incident ray and final reflected ray are anti-parallel?

(1) 90° (2) 60°
(3) 45° (4) 30°

2. A necessary condition for total internal reflection (TIR) is:

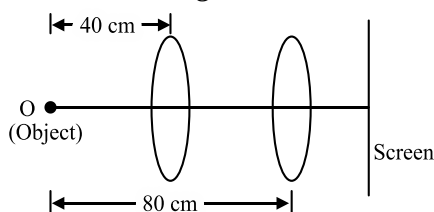
- (1) the angle of incidence in the rarer medium must be lesser than the critical angle for the two media.
(2) the angle of incidence in the denser medium must be greater than the critical angle for the two media.
(3) the angle of incidence in the denser medium must be lesser than the critical angle for the two media.
(4) the angle of incidence in the rarer medium must be greater than the critical angle for the two media.

3. Figure shows two plane mirrors parallel to each other and an object O placed between them. Then the distance of the first three images from the mirror M_2 will be: (in cm)



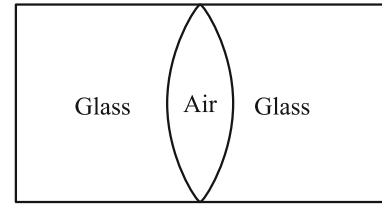
(1) 5, 10, 15 (2) 5, 15, 30
(3) 5, 25, 35 (4) 5, 15, 25

4. In displacement method, there are two positions of a lens for which we get clear image. First position of the lens is at 40 cm from object and second is at 80 cm, the focal length of lens is:



(1) 40 cm (2) $\frac{40}{3}$ cm
(3) 80 cm (4) $\frac{80}{3}$ cm

5. In the figure, an air lens of radii of curvature 10 cm ($R_1 = R_2 = 10$ cm) is cut in a cylinder of glass ($\mu = 1.5$). The focal length and the nature of the lens is:



(1) 15 cm, concave.
(2) 15 cm, convex.
(3) ∞ , neither concave nor convex.
(4) 0, concave.

6. The refractive index of the material of a prism is $\sqrt{2}$ and the angle of prism is 30° . One of its refracting faces is silvered. The incident beam of light will retrace back for angle of incidence:

(1) 0°
(2) 45°
(3) 60°
(4) 90°

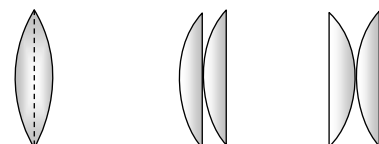
7. If the angle of minimum deviation is of 60° for an equilateral prism, then the refractive index of the material of the prism is:

(1) 1.41
(2) 1.5
(3) 1.6
(4) 1.73

8. Two plane mirrors are inclined at an angle of 60° with each other. A ray of light travelling horizontally is reflected first from one mirror and then from the other. The resultant deviation is:

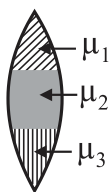
(1) 60° (2) 130°
(3) 180° (4) 240°

9. Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal lengths in three cases will be:



(1) 2 : 2 : 1 (2) 1 : 1 : 1
(3) 1 : 2 : 2 (4) 2 : 1 : 1

10. A convex lens is made up of three different materials as shown in the figure. For a point object placed on its axis, the number of images formed are:



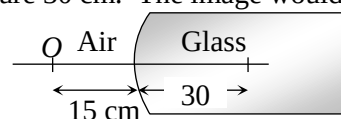
- (1) 1 (2) 5
(3) 4 (4) 3
11. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The refractive index of diamond is $\sqrt{6}$ and that of liquid is $\sqrt{3}$. If the light travels from diamond to the liquid, it will totally reflected when the angle of incidence is 30° .
Reason R: $\mu = \frac{1}{\sin C}$, where μ is the refractive index of diamond with respect to liquid.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

12. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The refractive index of a prism depends only on the kind of glass of which it is made of and the colour of light.
Reason R: The refractive index of a prism depends upon the refracting angle of the prism and the angle of minimum deviation.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

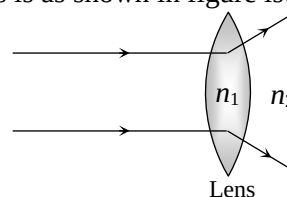
13. If an object is placed 10 cm in front of a concave mirror of focal length 20 cm the image will be:
(1) diminished, upright, virtual.
(2) enlarged, upright, virtual.
(3) diminished, inverted, real.
(4) enlarged, upright, real.

14. The radius of curvature for a convex lens is 40 cm, for each surface. Its refractive index is 1.5. The focal length will be:
(1) 40 cm
(2) 20 cm
(3) 80 cm
(4) 30 cm

15. A point object O is placed in front of a glass rod ($\mu_g = 1.5$) having spherical end of radius of curvature 30 cm. The image would be formed at:



- (1) 30 cm left.
(2) infinity.
(3) 1 cm to the right.
(4) 18 cm to the left.
16. A plane mirror is approaching you at a speed of 10 cm/sec. You can see your image in it. At what speed will your image approach you?
(1) 10 cm/sec
(2) 5 cm/sec
(3) 20 cm/sec
(4) 15 cm/sec
17. The relation between n_1 and n_2 , if behaviour of light rays is as shown in figure is:

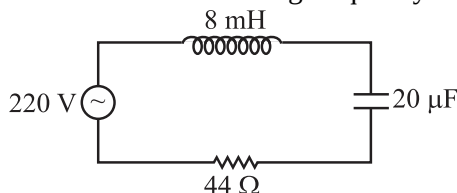


- (1) $n_1 \gg n_2$ (2) $n_2 > n_1$
(3) $n_1 > n_2$ (4) $n_1 = n_2$
18. A convex lens has a focal length f . It is cut into two parts along the dotted line as shown in the figure. The focal length of each part will be:



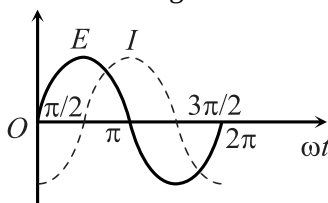
- (1) $\frac{f}{2}$ (2) f
(3) $\frac{3}{2}f$ (4) $2f$

19. For the series LCR circuit shown in the figure, what is the resonance frequency and the rms of the current at the resonating frequency:



- (1) $2500 \text{ rad} - \text{s}^{-1}$ and $5\sqrt{2} \text{ A}$
 (2) $2500 \text{ rad} - \text{s}^{-1}$ and 5 A
 (3) $2500 \text{ rad} - \text{s}^{-1}$ and $\frac{5}{\sqrt{2}} \text{ A}$
 (4) $25 \text{ rad} - \text{s}^{-1}$ and $5\sqrt{2} \text{ A}$
20. A virtual current of 4 A and 50 Hz flows in an ac circuit containing a coil. The power consumed in the coil is 240 W . If the virtual voltage across the coil is 100 V , its inductance will be:
- (1) $\frac{1}{3\pi} \text{ H}$ (2) $\frac{1}{5\pi} \text{ H}$
 (3) $\frac{1}{7\pi} \text{ H}$ (4) $\frac{1}{9\pi} \text{ H}$

21. The variation of the instantaneous current (I) and the instantaneous emf (E) in a circuit is as shown in figure.

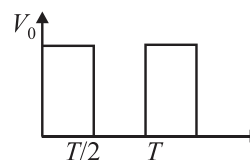


- Which of the following statements is **correct**?
- (1) The voltage lags behind the current by $\pi/2$.
 (2) The voltage leads the current by $\pi/2$.
 (3) The voltage and the current are in phase.
 (4) The voltage leads the current by π .
22. In an ac circuit, the instantaneous values of e.m.f. and current are $e = 200 \sin(314t)$ volt and $i = \sin\left(314t + \frac{\pi}{3}\right)$ ampere. The average power consumed in watt is:
- (1) 200 (2) 100
 (3) 50 (4) 25
23. Resonant frequency of a series LCR circuit is 600 Hz . Quality factor of the circuit is 4. Band width is:
- (1) 300 Hz
 (2) 150 Hz
 (3) 1200 Hz
 (4) 450 Hz

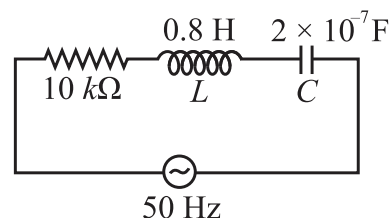
24. In a series resonant LCR circuit, the voltage across R is 100 volts and $R = 1 \text{ k}\Omega$ with $C = 2 \mu\text{F}$. The resonant frequency ω is 200 rad/s . At resonance the voltage across L is:

- (1) $4 \times 10^{-3} \text{ V}$ (2) $2.5 \times 10^{-2} \text{ V}$
 (3) 40 V (4) 250 V

25. The rms value of potential difference V shown in the figure is:

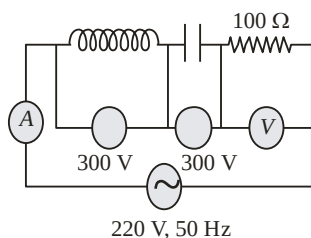


- (1) $\frac{V_0}{2}$
 (2) $\frac{V_0}{\sqrt{3}}$
 (3) V_0
 (4) $\frac{V_0}{\sqrt{2}}$
26. In an LCR series a.c. circuit, the voltage across each of the components, L , C and R is 50 V . The voltage across the LC combination will be:
- (1) 50 V (2) $50\sqrt{2} \text{ V}$
 (3) 100 V (4) 0 V (zero)
27. The Q -factor of LCR series circuit shown in the figure is:



- (1) 0.1 (2) 0.2
 (3) 0.3 (4) 0.4
28. In an LCR circuit, the energy is dissipated in:
- (1) R only
 (2) R and L only
 (3) R and C only
 (4) R , L and C
29. A coil has resistances 30Ω and inductive reactance 20Ω at 50 Hz frequency. If an ac source, of 200 V , 100 Hz is connected across the coil, the current in the coil will be:
- (1) $\frac{20}{\sqrt{13}} \text{ A}$
 (2) 2.0 A
 (3) 4.0 A
 (4) 8.0 A

30. In the circuit shown below, what will be the readings of the voltmeter and ammeter?



- (1) 800 V, 2 A (2) 200 V, 2 A
(3) 220 V, 2.2 A (4) 100 V, 2 A
31. The ratio of mean value over half cycle to r.m.s. value of A.C. is:
- (1) $2 : \pi$ (2) $2\sqrt{2} : \pi$
(3) $\sqrt{2} : \pi$ (4) $\sqrt{2} : 1$
32. For a series RLC circuit $R = X_L = 2X_C$. The impedance of the circuit and phase difference between V and i will be:
- (1) $\frac{\sqrt{5}R}{2}, \tan^{-1}(2)$
(2) $\frac{\sqrt{5}R}{2}, \tan^{-1}\left(\frac{1}{2}\right)$
(3) $\sqrt{5}X_C, \tan^{-1}(2)$
(4) $\sqrt{5}R, \tan^{-1}\left(\frac{1}{2}\right)$
33. An alternating current is given by the equation $i = i_1 \sin \omega t + i_2 \cos \omega t$. The rms current will be:
- (1) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$
(2) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$
(3) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
(4) $\frac{1}{\sqrt{2}}(i_1 + i_2)$
34. The phase difference between the alternating current and emf is $\pi/2$. Which of the following **cannot** be the constituent of the circuit?
- (1) LC (2) L alone
(3) C alone (4) R, L
35. A transformer having efficiency of 90% is working on 200 V and 3 kW power supply. If the current in the secondary coil is 6 A, the voltage across the secondary coil and the current in the primary coil respectively are:
- (1) 300 V, 15 A (2) 450 V, 15 A
(3) 450 V, 13.5 A (4) 600 V, 15 A

SECTION - B

36. A 220 volt input is supplied to a transformer. The output circuit draws a current of 2.0 ampere at 440 volts. If the efficiency of the transformer is 80%, the current drawn by the primary winding of the transformer is:
- (1) 3.6 ampere (2) 2.8 ampere
(3) 2.5 ampere (4) 5.0 ampere
37. The electromagnetic waves:
- (1) travel with the speed of sound
(2) travel with the same speed in all media
(3) travel in free space with the speed of light
(4) do not travel through a medium
38. Select the **correct** statement(s) from the following.
- A. Displacement current comes into play in a region where electric field is changing with time.
B. Displacement current $I_D = \epsilon_0 \frac{d\phi_E}{dt}$
C. In case of a steady electric flux linked with a region the displacement current is minimum.
Choose the most appropriate answer from the options given below.
- (1) A only (2) B only
(3) A and B only (4) A, B and C
39. Light wave is travelling along y -direction. If the corresponding $\vec{E}$ vector at any time is along the x -axis, the direction of $\vec{B}$ vector at that time is along :
-
- (1) y -axis (2) x -axis
(3) $+z$ -axis (4) $-z$ -axis
40. A plane electromagnetic wave travels in free space along x -axis. At a particular point in space, the electric field along y -axis is 9.3 Vm^{-1} . The magnetic induction (B) along z -axis is:
- (1) $3.1 \times 10^{-8} \text{ T}$ (2) $3 \times 10^{-5} \text{ T}$
(3) $3 \times 10^{-6} \text{ T}$ (4) $9.3 \times 10^{-6} \text{ T}$
41. The magnifying power of a telescope is 9. When it is adjusted for parallel rays, the distance between the objective and the eye piece is found to be 20 cm. The focal length of lenses are:
- (1) 18 cm, 2 cm (2) 11 cm, 9 cm
(3) 10 cm, 10 cm (4) 15 cm, 5 cm

42. If a source is transmitting electromagnetic wave of frequency 8.2×10^6 Hz, then wavelength of the electromagnetic waves transmitted from the source will be:

(1) 36.6 m (2) 40.5 m
(3) 42.3 m (4) 50.9 m

43. In the propagation of light waves, the angle between the plane of vibration and plane of polarisation is:

(1) 0° (2) 90°
(3) 45° (4) 80°

44. In the propagation of electromagnetic waves, the angle between the direction of propagation and plane of polarisation is:

(1) 0° (2) 45°
(3) 90° (4) 180°

45. Which of the following statements are true about the diffraction pattern?

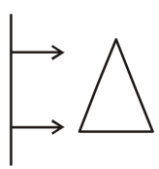
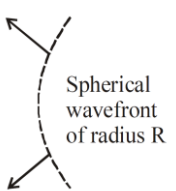
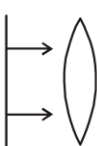
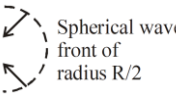
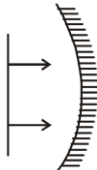
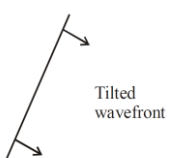
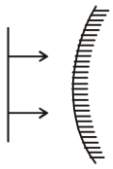
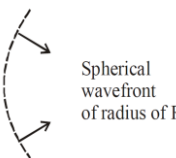
A. It has a central bright maximum of twice the width of other maxima.
B. The first null occurs at an angle $\lambda/2a$.
C. The intensity of maxima falls as we move away from the central maxima.
D. The bands are of decreasing width.

Choose the most appropriate answer from the options given below.

(1) B and C only (2) A and B only
(3) A, C and D only (4) A and C only

46. Match Plane wave incident on different surfaces in:

Match the **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

Choose the **correct** answer from the options given below.

(1) A-I, B-III, C-II, D-IV
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-II, C-I, D-III

47. The ratio of intensities at two points P and Q on a screen in young's double slit experiment when waves from sources S_1 and S_2 have phase difference of (a) 0° and (b) $\pi/2$ respectively, is:

(1) 1 : 4
(2) 4 : 1
(3) 1 : 2
(4) 2 : 1

48. In the Young's double slit experiment using a monochromatic light of wavelength λ , the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is:

(1) $(2n+1)\frac{\lambda}{2}$
(2) $(2n+1)\frac{\lambda}{4}$
(3) $(2n+1)\frac{\lambda}{8}$
(4) $(2n+1)\frac{\lambda}{16}$

49. In Young's double slit expt. the distance between two coherent sources is 0.1 mm. The distance of the screen from the source is 20 cm. Wavelength of light used is 5460 Å. The angular position of the first dark fringe is:

(1) 0.08°
(2) 0.16°
(3) 0.20°
(4) 0.32°

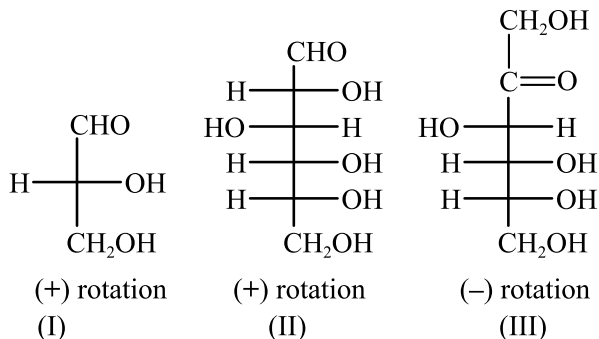
50. The Fraunhofer 'diffraction' pattern of a single slit is formed in the focal plane of a lens of focal length 1 m. The width of slit is 0.3 mm. If third minimum is formed at a distance of 5 mm from central maximum, then wavelength of light will be:

(1) 5000 Å (2) 2500 Å
(3) 7500 Å (4) 8500 Å

SECTION-(II) CHEMISTRY

SECTION - A

51. Optical rotations of some compounds along with their structures are given below, which of them have D configuration?



- (1) I, II, III
(2) II, III only
(3) I, II only
(4) III only
52. Choose the **correct** relationship for glucose and fructose:
- (1) these are functional isomers
(2) these are chain isomers
(3) these are position isomers
(4) None of these
53. Which amino acid contains sulfur in its structure?
- (1) Alanine
(2) Glycine
(3) Methionine
(4) Valine
54. What is the relation between the two cyclic hemiacetal form of glucose?
- (1) They are anomers
(2) They are isomers
(3) They are epimer
(4) They are pyranose
55. Which reagent is used to convert glucose into saccharic acid?
- (1) $\text{Br}_2/\text{H}_2\text{O}$
(2) Nitric acid
(3) Alkaline solution of iodine
(4) Ammonium hydroxide

56. Which of the statement about "Denaturation" given below are **correct**?

- (A) Denaturation of proteins causes loss of secondary and tertiary structures of the protein.
(B) Denaturation leads to the conversion of double strand of DNA into single strand.
(C) Denaturation affects primary structure which gets distorted.
- (1) (B) and (C) only
(2) (A) and (C) only
(3) (A) and (B) only
(4) (A), (B) and (C)

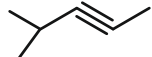
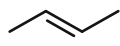
57. Phosphodiester linkage is present between which carbon atoms of pentose sugars of nucleotides?

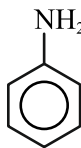
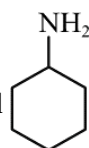
- (1) 5' and 3'
(2) 1' and 5'
(3) 5' and 5'
(4) 3' and 3'

58. Which of the following is a protein deficiency disease?

- (1) Rickets
(2) Pyorrhea
(3) Anaemia
(4) Beri-Beri

59. Which of the following will give white precipitate with ammoniacal silver nitrate solution?

- (1) $\text{CH}_3 - \text{C} \equiv \text{CH} - \text{CH}_3$
(2) 
(3) 
(4) $\text{Ph} - \text{C} \equiv \text{CH}$

60.  and  can be differentiated by:

- (1) carbylamine reaction
(2) H_2SO_4
(3) diazotisation followed by β -naphthol
(4) mustard oil reaction

61. Which of the following is **true**?

- (1) Tollen's reagent gives a positive test with all aldehyde.
- (2) Fehling's solution gives a positive test with all aldehyde.
- (3) Tollen's reagent gives a positive test with all carboxylic acid.
- (4) Tollen's reagent gives a positive test with α -methyl ketone.

62. Which of the following compounds does **not** give reaction with acidic KMnO_4 ?

- (1)
- (2)
- (3)
- (4)

63. Which of the following compound can **not** produce CO_2 on reaction with NaHCO_3 ?

- (1)
- (2)
- (3)
- (4)

64. Which of following complex has higher Δ_o value?

- (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- (3) $[\text{Fe}(\text{CN})_6]^{3-}$
- (4) All have equal Δ_o Values

65. When AgNO_3 is added to a solution of $\text{CO}(\text{NH}_3)_5\text{Cl}_3$, the precipitate of AgCl shows two ionized chloride ions. This means that:

- (1) two chlorine atoms satisfy primary valency and one secondary valency.
- (2) one chlorine atom satisfies primary as well as secondary valency.
- (3) three chlorine atoms satisfy primary valency.
- (4) three chlorine atoms satisfy secondary valency.

66. Given below are two statements:

Statement I: In coordination compounds metals atom shows two types of linkages (valencies)-primary and secondary.

Statement II: The secondary valency is ionisable. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

67. The complex ion which has no 'd'-electrons in the central metal ion is:

- (1) $[\text{MnO}_4]^-$
- (2) $[\text{Co}(\text{NH}_3)_6]^{3+}$
- (3) $[\text{Fe}(\text{CN})_6]^{3-}$
- (4) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

68. Which one of the following square planar complexes exist in cis and trans isomeric forms ?

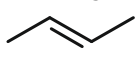
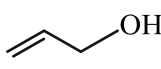
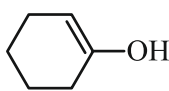
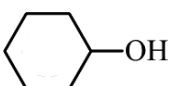
- (1) Ma_4
- (2) Ma_3b
- (3) Ma_2b_2
- (4) Mabcd

69. and can be differentiated by:

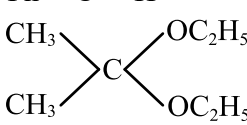
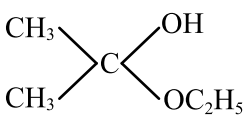
- (1) NaOH
- (2) Na metal

- (3) NaHCO_3 (4) FeCl_3
70. $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$ and $\text{Ph}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$ can be differentiated by:
- (1) Tollen's reagent
 - (2) Fehling's solution
 - (3) Lucas reagent
 - (4) Victor meyer's test

71. Which of the following compounds decolourise $\text{Br}_2-\text{H}_2\text{O}$ and also give positive test with neutral FeCl_3 :

- (1) 
- (2) 
- (3) 
- (4) 

72. Which of the following compounds give positive test with Tollen's reagent?

- (1) $\text{Ph}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$
- (2) 
- (3) 
- (4) $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$

73. Which has maximum paramagnetic nature?

- (1) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
- (2) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- (3) $[\text{Fe}(\text{CN})_6]^{4-}$
- (4) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$

74. The **correct** statements among I, II and III are:
- (I) Valence bond theory cannot explain the color exhibited by transition metal complexes.
 - (II) Valence bond theory can predict quantitatively the magnetic properties of transition metal complexes.

(III) Valence bond theory cannot distinguish ligands as weak and strong field ones.

- (1) (I) and (II) only
- (2) (I), (II) and (III)
- (3) (I) and (III) only
- (4) (II) and (III) only

75. How many geometrical isomers are possible for the complex ion $[\text{Cr}(\text{NH}_3)\text{Cl}_3(\text{OH})_2]^{2-}$?

- (1) 3
- (2) 4
- (3) 5
- (4) None of these

76. How many ions are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?

- (1) 6
- (2) 4
- (3) 3
- (4) 2

77. The **correct** charge and coordination number of 'Fe' in $\text{K}_3[\text{Fe}(\text{CN})_6]$ is

- (1) +2, 4
- (2) +3, 6
- (3) +2, 6
- (4) +3, 3

78. Spin only magnetic moment 2.84 BM is given by: (Atomic number of Ni = 28, Ti = 22, Cr = 24 and Co = 27)

- (1) Ni^{2+}
- (2) Ti^{3+}
- (3) Cr^{2+}
- (4) Co^{2+}

79. How does the ionic radius with same charge trend within a transition series typically behave?

- (1) Increases with atomic number
- (2) Decreases with atomic number
- (3) Remains constant
- (4) Varies irregularly

80. The element showing variable oxidation states of +2, +3, +4, +6 and +7 is:

- (1) Cr
- (2) Mn
- (3) Co
- (4) V

81. The general electronic configuration of 5d series elements is:

- (1) $[\text{Kr}]4f^{14}5d^{1-10}6s^{0-2}$
- (2) $[\text{Xe}]4f^{1-14}5d^{0-10}6s^{0-2}$
- (3) $[\text{Xe}]4f^{0-14}5d^{1-10}6s^{1-2}$
- (4) $[\text{Rn}]5f^{0-14}6d^{1-10}7s^2$

82. Transition elements act as catalyst because:

- (1) their melting points are high.
- (2) their I.P. values are high.
- (3) they have high density.
- (4) they can show variable oxidation states.

83. The most characteristic oxidation state of lanthanides is:

- (1) +2
- (2) +3
- (3) +4
- (4) None of these

84. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Cu^{2+} in water is more stable than Cu^+ .

Reason (R): Enthalpy of hydration for Cu^{2+} is much less than that of Cu^+ .

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

85. $\text{Cr}_2\text{O}_7^{2-} + \text{X} \xrightarrow{\text{H}^+} \text{Cr}^{3+} + \text{H}_2\text{O} + \text{Oxidized product of X}$. X in the above reaction cannot be

- (1) $\text{C}_2\text{O}_4^{2-}$
- (2) Fe^{2+}
- (3) SO_4^{2-}
- (4) S^{2-}

SECTION - B

86. What is the percentage of lanthanoid metal in Mischmetal?

- (1) 90%
- (2) 20%
- (3) 5%
- (4) 95%

87. Which of the following is arranged in order of increasing melting point?

- (1) $\text{Zn} < \text{Cu} < \text{Ni} < \text{Fe}$
- (2) $\text{Fe} < \text{Ni} < \text{Cu} < \text{Zn}$
- (3) $\text{Ni} < \text{Fe} < \text{Zn} < \text{Cu}$
- (4) $\text{Cu} < \text{Zn} < \text{Fe} < \text{Ni}$

88. Although +3 is the characteristic oxidation state for lanthanoids but cerium also shows +4 oxidation state because _____.

- (1) It has variable ionisation enthalpy.
- (2) It has a tendency to attain noble gas configuration.
- (3) It has a tendency to attain f^1 configuration.
- (4) It resembles Pb^{4+} .

89. What colouration in Schiff's test indicates the presence of aldehyde?

- (1) Blue
- (2) Pink
- (3) Grey
- (4) Orange

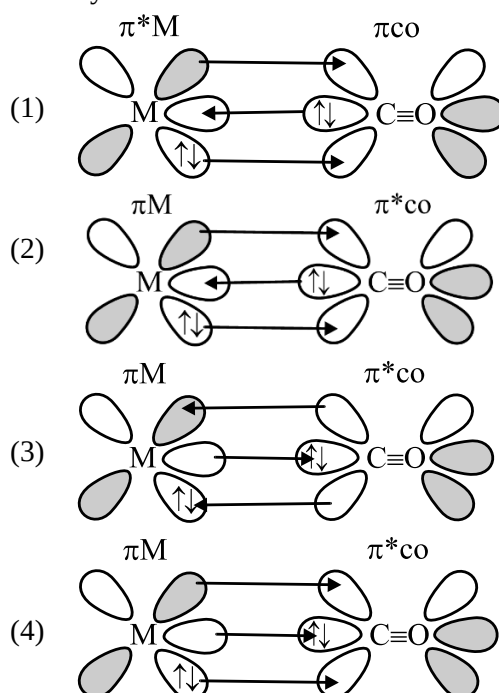
90. What is colour of precipitate formed in iodoform test?

- (1) Grey
- (2) Yellow
- (3) Black
- (4) Blue

91. Lassaigne's test is **not** used for the detection of:

- (1) N
- (2) S
- (3) Cl
- (4) O

92. Metal 'M' form a carbonyl compound in which it is present lower valance state. Which of the following bonding is possible in this metal carbonyl?



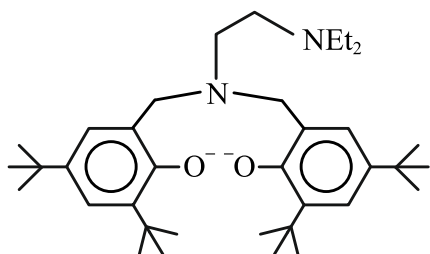
93. The IUPAC name of the complex $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ is:

- (1) Aquatetramminechloridocobalt (III) chloride
- (2) Chloridoaquatetramminechloridocobalt (III) chloride
- (3) Chloridoaquatetramminechloridocobalt (II) chloride
- (4) Tetraammineaquachloridocobalt (III) chloride

94. Which of the following is a π -acid ligand?

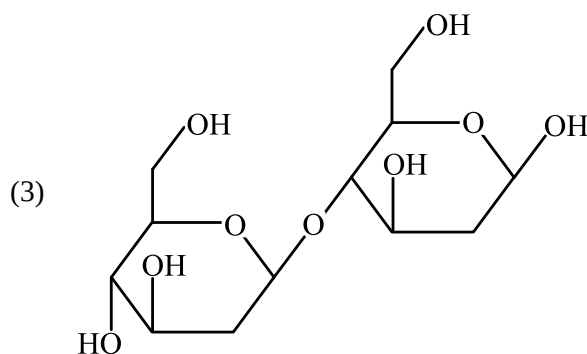
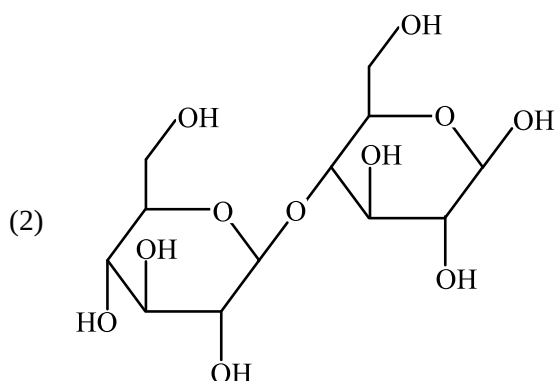
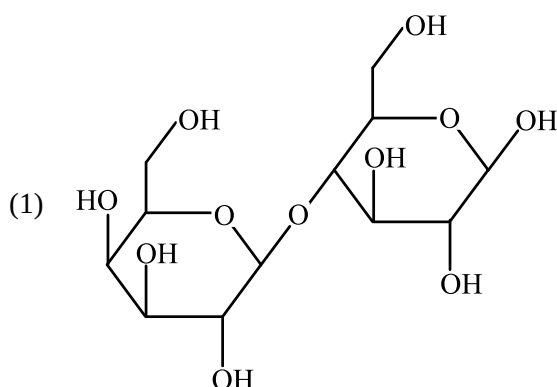
- (1) NH_3
- (2) CO
- (3) F^-
- (4) $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$

95. The following ligand is



- (1) Hexadentate
- (2) Tetradentate
- (3) Bidentate
- (4) Tridentate

96. Find out the structure of lactose



- (4) None of these

97. The beta and alpha glucose have different specific rotations. When either is dissolved in water, their rotation changes until the same fixed value results. This is called

- (1) epimerisation
- (2) racemisation
- (3) anomerisation
- (4) mutarotation

98. Which of the following statements regarding a peptide linkage in a protein molecule is **incorrect**?

- (1) It is an amide linkage.
- (2) It has partial double bond character.
- (3) It is hydrophobic in nature.
- (4) It connects polyamide polymeric chains through hydrogen bonding.

99. In cellulose, D-glucose units are joined by:

- (1) α -1, 4-glycosidic linkage.
- (2) β -1, 6-glycosidic linkage.
- (3) β -1, 4-glycosidic linkage.
- (4) peptide linkage.

100. α -Helix is a secondary structure of proteins formed by twisting of polypeptide chain into right-handed screw like structures. Which type of interactions are responsible for making the α -helix structure stable?

- (1) London-Dispersion forces
- (2) Ionic bond
- (3) H-bonding
- (4) Disulphide linkages

SECTION-(III) BOTANY**SECTION - A**

101. Read the following statement regarding lactic acid bacteria (LAB).

- A. *Lactobacillus* is called LAB.
- B. LAB improves milk nutritional quality by increasing vitamin B₆.
- C. LAB increases disease causing microbes in gut.
- D. LAB is used to prepare curd.

Choose the most appropriate answer from the options given below:

- (1) A, C and D only
- (2) A, B and C only
- (3) B, C and D only
- (4) A and D only

102. In dog flower, a pink flowered plant is test crossed. What is the phenotypical probability in the progeny?

- (1) 1 Red : 2 Pink : 1 White
- (2) 1 Red : 2 Pink
- (3) 1 Pink : 1 White
- (4) 3 Red : 1 Pink

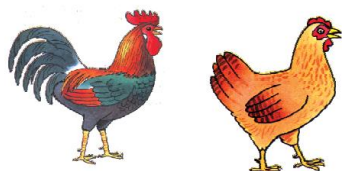
103. If three genes control skin colour in human with the dominant forms A, B and C and the recessive forms a, b and c. The lightest skin colour will be shown by the genotype:

- (1) AABBCC. (2) AaBbCc.
- (3) AABbCc. (4) aabbcc.

104. The number of genotypes and phenotypes in ABO blood groups are:

- (1) one and four respectively.
- (2) four and six respectively.
- (3) six and four respectively.
- (4) two and six respectively.

105. Select the **correct** option for the following diagram P and Q.



(P)

(Q)

- (1) P is female and has a pair of dissimilar chromosomes ZW and Q is male and has two similar ZZ chromosomes.
- (2) P is male and has a pair of similar chromosomes ZZ and Q is female and has two dissimilar ZW chromosomes.
- (3) P is female and Q is male and has a pair of dissimilar chromosomes ZW.
- (4) P is female and Q is male and has similar ZZ chromosomes.

106. Mendelian disorders are mainly determined by alteration or mutation in the:

- (1) two genes. (2) single gene.
- (3) four genes. (4) three genes.

107. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types of nucleic acids found in living systems.

Reason R: DNA acts as the genetic material in most of the organisms.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

108. The medium for growing bacteria and many fungi in the laboratory is called:

- (1) antibiotic.
- (2) baking media.
- (3) fermentation media.
- (4) nutritive culture media.

109. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Trichoderma polysporum</i>	(I)	Cyclosporin A
(B)	<i>Monascus purpureus</i>	(II)	Statins
(C)	<i>Streptococcus</i>	(III)	Streptokinase
(D)	<i>Clostridium butylicum</i>	(IV)	Butyric acid

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-II, C-III, D-IV

110. Which of the following fix nitrogen in free living form?

- (1) *Rhizobium*
- (2) *Azospirillum*
- (3) *Volvox*
- (4) *Rhizopus*

111. Primary treatment of sewage waste involves which processes?
- (1) Filtration only
 - (2) Sedimentation only
 - (3) Filtration and sedimentation only
 - (4) Filtration, sedimentation and microbial proliferation

112. Identify from the following microorganisms which help in absorption of phosphorus from soil by plants.

A. *Rhizobium*
 B. *Frankia*
 C. *Anabaena*
 D. *Glomus*

Choose the most appropriate answer from the options given below:

- (1) A, B and C
- (2) A and C
- (3) A and B
- (4) D only

113. Ganga and Yamuna action plan is initiated by:

- (1) ministry of environment and forest.
- (2) ministry of agriculture.
- (3) ministry of wild-life conservation.
- (4) Khadi and Village Industries Commission.

114. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: RNA is known to be catalytic and less reactive in comparison to DNA.

Reason R: RNA functions as adapter molecule.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

115. Match List-I with List-II.

List-I		List-II	
(A)	<i>Aspergillus niger</i>	(I)	Lactic acid
(B)	<i>Acetobacter aceti</i>	(II)	Ethanol
(C)	<i>Saccharomyces cerevisiae</i>	(III)	Acetic acid
(D)	<i>Lactobacillus</i>	(IV)	Citric acid

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-I, C-IV, D-III

116. Satellite DNA is important because it:

- (1) is same in all members of the population.
- (2) codes for enzymes needed for DNA replication.
- (3) codes for proteins needed in cell cycle.
- (4) shows high degree of polymorphism in population.

117. Read the following gases and select the **correct** option containing gases present in biogas.

A. Methane
 B. Hydrogen
 C. Carbon dioxide
 D. Sulphur dioxide

Choose the most appropriate answer from the options given below:

- (1) A, C and D only
- (2) A, B and C only
- (3) B, C and D only
- (4) A and B only

118. Out of 64 codons, 61 code for 20 types of amino acids. This property of genetic code is called as:

- (1) all codons are same.
- (2) degeneracy of genetic code.
- (3) universality of codons.
- (4) unambiguous nature of code.

119. ___A___ is present in DNA only and ___B___ is present in RNA only. Identify **A** and **B** respectively.

A	B
(1) Cytosine	thymine
(2) Thymine	uracil
(3) Uracil	thymine
(4) Thymine	cytosine

120. Which one of the following is **not** true about antibiotics?

- (1) First antibiotic was discovered by Alexander Flemming.
- (2) Full potential as an effective antibiotic was established by Fleming, Ernest Chain and Howard Florey.
- (3) Fleming, Chain and Florey were awarded the Nobel Prize in 1946 for the discovery of penicillin.
- (4) Penicillin was the first antibiotic to be discovered.

121. The supernatant left after sedimentation of sewage water during primary treatment is known as:

- (1) effluent.
- (2) flocs.
- (3) sludge.
- (4) biogas.

122. Match List-I with List-II.

List-I		List-II	
(A)	Bacteriophage $\phi \times 174$	(I)	5386 nucleotides
(B)	Bacteriophage lambda	(II)	48502 bp
(C)	<i>Escherichia coli</i>	(III)	4.6×10^6 bp
(D)	Haploid content of human DNA	(IV)	3.3×10^9 bp

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-I, C-III, D-IV
- (3) A-I, B-II, C-IV, D-III
- (4) A-I, B-II, C-III, D-IV

123. Select the **correct statements about Baculoviruses (*Nucleopolyhedrovirus*).**

- A. Host specificity.
- B. Narrow spectrum applications.
- C. Effects on non-target pathogens.
- D. Utility in IPM programme.

Choose the most appropriate answer from the options given below:

- (1) A, B and D only
- (2) B and C only
- (3) C only
- (4) A, B, C and D

124. Find out the **incorrect option.**

- (1) Euchromatin is transcriptionally active part of chromatin.
- (2) Heterochromatin is densely packed.
- (3) Euchromatin gets light stained.
- (4) Histones are scattered in cytoplasm in eukaryotes.

125. If the sequence of bases in one strand of DNA is ATGCAAGGA, what would be the sequence of bases on complementary strand?

- (1) TTGCATGCA
- (2) UUGCAUGCA
- (3) TACGTCCT
- (4) AACGUACGU

126. High biological oxygen demand in a waste water body means:

- (1) water is not polluted.
- (2) water is polluted.
- (3) water body contains only solid materials.
- (4) Both (2) and (3).

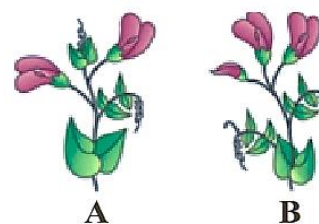
127. Select the **incorrect statement/s given below.**

- A. Mendel proposed two general rules to consolidate his understanding of inheritance in dihybrid crosses.
- B. Mendel self-pollinated the F_1 plants.
- C. The symbol σ is used to denote the female (eggs).
- D. The law of segregation is based on the fact that the alleles show blending.

Choose the most appropriate answer from the options given below:

- (1) A only
- (2) B and C only
- (3) D only
- (4) A, C and D only

128. Identify the position of flowers in the given images A and B.



- (1) A- Axial, B- Axial
- (2) A- Axial, B- Terminal
- (3) A- Terminal, B- Axial
- (4) A- Terminal, B- Terminal

129. In Mendelian dihybrid cross, when pea plant heterozygous for round and yellow seeds are self-crossed, what would be the genotype of round and green seeds obtained in F_2 generation:

- (1) $RrYy$ and $RrYY$.
- (2) $Rryy$ and $rryy$.
- (3) $rryy$ and $rrYY$.
- (4) $Rryy$ and $RRyy$.

130. If alleles B and b control the size and shape of starch grains in pea plant. Which of the given character will be expressed by genotypic combination Bb?

- (1) Round and large size
- (2) Wrinkle and intermediate size
- (3) Round and small size
- (4) Round and intermediate size

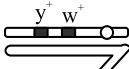
131. A child has O blood group and his mother has O blood group. The genotype of the father's blood group can **not** be:

- (1) ii .
- (2) $I^B i$.
- (3) $I^A I^B$.
- (4) $I^A i$.

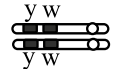
132. Each antibiotic is effective only against:

- (1) insects.
- (2) aphids.
- (3) butterfly caterpillars.
- (4) bacteria.

133. Read the following and select the **correct** statement.

- (1)  represents dominant recessive type genotype in pea.

- (2) A cross between plants differing in one paired trait is called monohybrid cross.

- (3)  represents male with recessive Yellow, white linked genes in pea.

- (4) Chromosomes as well as genes occur in pairs in diploid organisms.

134. The percentage of ab gamete produced by AaBb parent will be:

- (1) 25%.
- (2) 50%.
- (3) 75%.
- (4) 12.5%.

135. Given below are two statements:

Statement I: Biocontrol refers to the use of biological methods for controlling plant diseases and pests.

Statement II: In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

SECTION - B

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams.

Reason R: The number of sewage treatment plants has not increased enough to treat sewage so sewage is often discharged directly into rivers leading to their pollution and increase in water-born diseases.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

137. Colour blindness is a/an:

- (1) autosomal recessive disorder.
- (2) autosomal dominant disorder.
- (3) X-linked recessive disorder.
- (4) Y-linked recessive disorder.

138. In the process of transcription in eukaryotes, the RNA polymerase-III transcribes:

- (1) tRNA, 5SrRNA and snRNA.
- (2) rRNAs-28 S, 18 S and 5.8 S.
- (3) precursor of mRNA, hnRNA.
- (4) tRNA, 5.8S rRNA and snRNA.

139. How many genotypes and phenotypes are possible of the children if the genotypes of husband and wife are $I^A I^B$ and $I^A i$ respectively?

- (1) 3 genotypes; 4 phenotypes
- (2) 4 genotypes; 3 phenotypes
- (3) 4 genotypes; 4 phenotypes
- (4) 3 genotypes; 3 phenotypes

140. Which of the following can be used in patients who have undergone myocardial infarction:

- (1) streptokinase.
- (2) lipase.
- (3) cyclosporin A.
- (4) cholesterol.

141. Which of the following is **not** an example of recessive gene?

- (1) Green seed in pea plant
- (2) White flower in pea plant
- (3) Red flower in snapdragon
- (4) Wrinkled seed in pea plant

142. Select the statement which best describes the relationship between gene linkage and recombination.

- (1) Tightly linked genes show higher recombination frequencies.
- (2) Genes located on different chromosomes show high recombination.
- (3) Genes located on the same chromosome tend to be inherited independently.
- (4) Tightly linked genes show lower recombination frequencies.

143. If coding strand of DNA contains 10 thymidylic acids and 5 adenylic acids, the number of thymine residue and adenine residue in mRNA will be:

- (1) 5, 10.
- (2) 0, 5.
- (3) 10, 5.
- (4) 10, 10.

144. Select the **correct** statements related to Bt spores used as biocontrol agent.

- A. Available as sachets.
- B. In the lungs of the larvae, the toxin is released and the larvae get killed.
- C. Mixed with organic solvent.
- D. Mixed with water.

Choose the most appropriate answer from the options given below:

- (1) A and D only
- (2) A, B and C only
- (3) C only
- (4) A, B, C and D

145. Select the **correct** match from the following for flower colour in garden pea.

- (1) Dominant- White
- (2) Recessive- Violet
- (3) Genotype-Vv for Violet
- (4) Phenotype-White for VV

146. Given below are two statements:

Statement I: Toddy, a traditional drink is made by fermenting sap from palms.

Statement II: Microbes are used to ferment bamboo shoots to make foods.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

147. Select the **correct** sequence of Alfred Hershey and Martha Chase experiment for finding out the genetic material.

- (1) Centrifugation → Infection → Blending
- (2) Infection → Blending → Centrifugation
- (3) Blending → Infection → Centrifugation
- (4) Infection → Centrifugation → Blending

148. Expressed Sequence Tags (ESTs) referred to as:

- (1) all genes that are expressed as proteins.
- (2) all genes whether expressed or unexpressed.
- (3) certain important expressed genes.
- (4) all genes that are expressed as RNA.

149. The enzyme DNA ligase is used:

- (1) for replication fork formation.
- (2) to start DNA replication.
- (3) in translation.
- (4) to join the discontinuously synthesised fragments during replication.

150. 'Swiss cheese' bears large holes due to the production of CO₂ by which microbe?

- (1) *Lactobacillus*
- (2) *Saccharomyces cerevisiae*
- (3) *Propionibacterium sharmanii*
- (4) *Aspergillus niger*

SECTION-(IV) ZOOLOGY

SECTION - A

151. Study of abnormalities by taken out the amniotic fluid of embryo is called:

- (1) MTP. (2) Amniocentesis.
(3) MRI. (4) Infertility.

152. Match List-I with List-II.

List-I		List-II	
(A)	The Pill	(I)	Semen contains no sperms
(B)	Vasectomy	(II)	Prevents ovulation
(C)	Condom	(III)	Reduce fertility capacity of sperms
(D)	Copper T	(IV)	Prevents sperms reaching cervix

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

153. Choose the **correct** option about test tube baby.

- (1) Fertilisation occur inside the female genital tract and growth in test tube.
(2) Rearing of a prematurely born baby in an incubator.
(3) Fertilisation take place outside and gestation inside the womb of mother.
(4) Both fertilisation and development occur outside the female genital tract.

154. Which of the following is **not** a part of the first line defenses of our body?

- (1) Acidic pH
(2) Saliva
(3) Antibodies
(4) Mucus membrane of urinogenital tract

155. Given below are two statements.

Statement I: The government was forced to take up serious measures to check population growth rate.

Statement II: An ideal contraceptive should be user-friendly.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
(2) Statement I is incorrect, but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

156. Which of the following is **not** legitimate for reducing birth rate?

- (1) Ban on marriages.
(2) Use of IUDs.
(3) Use of contraceptive pills.
(4) Late marriages.

157. Read the following statements (A-D).

- A. Medical Termination of Pregnancy during the first trimester is generally safe.
B. Generally chances of conception are nil until mother breast-feeds the infant up to two years.
C. Intrauterine devices like copper T are effective contraceptives.
D. Contraception pills may be taken up to one week after coitus to prevent conception.

Which of the following statements are **incorrect**?

- (1) B and C only
(2) C and D only
(3) A and D only
(4) B and D only

158. Antibodies in our body are complex:

- (1) lipoproteins.
(2) steroids.
(3) prostaglandins.
(4) proteins.

159. Most important component of oral contraceptive pill is:

- (1) thyroxine.
(2) LH.
(3) progesterone.
(4) FSH.

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Progestogens alone or in combination with estrogen can also be used by females as implants.

Reason R: Implants are used by both males and females.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

161. Which of the following is/are well known plants with hallucinogenic properties?

- (1) *Papaver somniferum*
- (2) *Atropa belladonna*
- (3) *Erythroxylum coca*
- (4) All of these

162. *In-vitro* fertilisation is a technique that involves transfer of which of the following into the uterus?

- (1) Zygote only.
- (2) Embryo only up to 6 cell stage.
- (3) Early embryo more than 8 cell stage.
- (4) Embryo of 2 cell stage only.

163. Which of the following is the most widely accepted method of contraception in India, as at present?

- (1) Cervical caps (2) Tubectomy
- (3) Diaphragms (4) IUDs

164. Artificial insemination means:

- (1) introduction of sperms of a healthy donor directly into the ovary.
- (2) transfer of sperm from a healthy donor to a test tube containing ova.
- (3) transfer of sperm from only husband to a test tube containing ova.
- (4) artificial introduction of sperms of a healthy donor into the vagina.

165. Given below are two statements.

Statement I: IUDs are ideal contraceptives for the females who want to delay pregnancy.

Statement II: Pills have to be taken daily for a period of 21 days starting preferably within the first five days of the menstrual cycle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Which of the following is/are **incorrect** regarding antiallergic chemical?

- (1) Adrenaline (2) Serotonin
- (3) Anti-histamine (4) Steroids

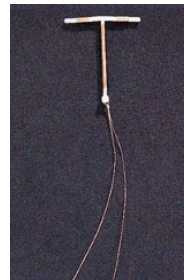
167. Tubectomy is a method of sterilisation in which:

- (1) a small part of the fallopian tube is removed or tied up.
- (2) ovaries are removed surgically.
- (3) a small part of vas deferens is removed or tied up.
- (4) uterus is removed surgically.

168. Which of the following is /are biological response modifiers?

- (1) Insulin (2) Alpha interferon
- (3) Serotonin (4) Adrenaline

169. Identify the diagram given below and choose the **correct** one from the following option.



- (1) Vaults (2) Implants
- (3) Cu T (4) Lippes loop

170. Read the following statements (A-E).

- A. Saheli is a non-steroidal pill.
- B. Saheli is a once a week pill.
- C. Insertion of IUDs within 130 hours of coitus prevents conception.
- D. The Government of India legalised MTP in 1951.
- E. Nearly 5 to 10 million MTPs are performed in a year all over the world.

How many of the above statements is/are **correct**?

- (1) Two
- (2) One
- (3) Four
- (4) Three

171. The cell-mediated immunity inside the human body is carried out mainly by:

- (1) Erythrocytes.
- (2) T-lymphocytes.
- (3) B-lymphocytes.
- (4) Thrombocytes.

172. Read the following statements (A-D).

- A. Barbiturates cause relaxation and temporary euphoria.
- B. Hashish is derived from *Papaver somniferum*.
- C. Opium stimulates the nervous system and causes hallucinations.
- D. Morphine leads to delusions but not disturbed emotions.

Which of the following statements are **correct**?

- (1) B and C only
- (2) C and D only
- (3) A and C only
- (4) A and B only

173. Which of the following is a **correct** statement?

- (1) IUDs once inserted need not be replaced.
- (2) IUDs are generally inserted by the user herself.
- (3) IUDs increase phagocytosis of sperms in the uterus.
- (4) IUDs suppress gametogenesis.

174. In which of the following techniques, the embryos are transferred to assist those females who cannot conceive?

- (1) GIFT and ICSI
- (2) ZIFT and IUT
- (3) GIFT and AI
- (4) ICSI and ZIFT

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	HIV	(I)	Entry of sperm through cervix is blocked
(B)	Spermicidal cream	(II)	Sperms are directly injected into the ovum
(C)	ICSI	(III)	Retrovirus
(D)	Vaults	(IV)	Increase contraceptive efficiency

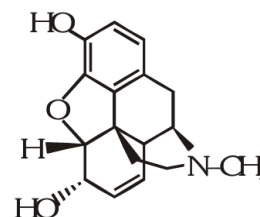
Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-I, C-IV, D-II

176. Which of the following scientist disproved the good humor hypothesis of health?

- (1) Charles Darwin
- (2) Robert Brown
- (3) William Bateson
- (4) William Harvey

177. Identify the diagram given below and choose the **correct** one from the following given options?



- (1) Serotonin
- (2) Haemoglobin
- (3) Lecithin
- (4) Morphine

178. Which of the following is/are **incorrect** regarding the function of nicotine?

- (1) It stimulates adrenal gland.
- (2) It is an alkaloid.
- (3) It is only present in coffee.
- (4) It causes cancer.

179. Heroin, commonly called smack, is chemically X, which is white, odourless, bitter crystalline compound, obtained by Y of morphine.

Choose the option which fills the blank **correctly**.

- (1) X-diacetylmorphine, Y-methylation
- (2) X-diacetylmorphine, Y-acetylation
- (3) X-benzodiazepines, Y-amination
- (4) X-amphetamines, Y-acetylation

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Injection of dead or inactivated pathogens causes passive immunity.

Reason R: Passive immunity does not provide a permanent cure for disease.

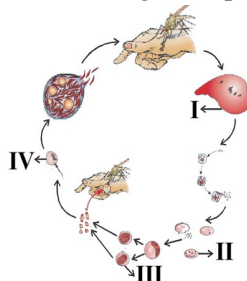
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

181. Choose the **odd** one out w.r.t. secondary lymphoid organs.

- (1) Thymus (2) Spleen
- (3) Appendix (4) Tonsils

182. Identify the stages marked as **I, II, III** and **IV** in the life cycle of *Plasmodium* and choose the **incorrect** one from given options.



- (1) Sporozoites in the liver.
- (2) Sexual reproduction of *Plasmodium* in RBCs.
- (3) Gametocytes in WBCs.
- (4) Fertilisation of gametocytes in mosquito's gut

183. The zygote or early embryos with upto 8 blastomeres could then be transferred into the:

- (1) ovaries.
- (2) fallopian tube.
- (3) uterus.
- (4) cervix.

184. Given below are two statements.

Statement I: Health means only absence of disease or physical fitness.

Statement II: The substances by which exaggerated immune response is produced are called allergens.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

185. Which of the following is/are **not** symptoms of allergy?

- (1) Sneezing
- (2) Watery eyes
- (3) Running nose
- (4) Easy breathing

SECTION - B

186. A contraceptive pill prevents ovulation by:

- (1) blocking the fallopian tube.
- (2) inhibiting release of FSH & LH.
- (3) stimulating release of FSH & LH.
- (4) causing immediate degeneration of released ovum.

187. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Cancer causing viruses called oncogenic viruses modify the host's genome.

Reason R: Cancer detection is based on biopsy, bone marrow tests and histopathological studies of the tissue and blood.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

188. HIV that causes AIDS, first starts destroying:

- (1) B-lymphocytes.
- (2) leucocytes.
- (3) thrombocytes.
- (4) helper T-lymphocytes.

189. Which of the following is/are popular brands for male condoms?

- (1) Nirodh (2) Unwanted 72
- (3) LNG-20 (4) Multiload 375

190. Given below are two statements.

Statement I: STIs are a major threat to a healthy society.

Statement II: Incidences of STIs are reported to be very high among persons in the age group of 15-24 years.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

191. Which of the following sets of diseases is caused by bacteria?

- (1) Tetanus and mumps
- (2) Herpes and influenza
- (3) Cholera and tetanus
- (4) Typhoid and smallpox

192. Read the following statements (A-D).

- A. Sporozoite stage infects human body.
- B. Gametocytes multiply by sexual method of reproduction in the human liver.
- C. Human RBCs are ruptured and released heparin.
- D. Haemozoin causes high recurring fever.

Which of the following statements are **incorrect** regarding malaria?

- (1) B and C only
- (2) C and D only
- (3) A and D only
- (4) A and B only

193. Which of the following is a hormone releasing Intra Uterine Device (IUD)?

- (1) Multiload 375
- (2) LNG – 20
- (3) Cervical cap
- (4) Vault

194. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Allergy	(I)	Typhoid fever
(B)	Polio	(II)	IgE
(C)	Widal test	(III)	Non-infectious
(D)	ELISA test	(IV)	AIDS

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-IV, D-II

195. Grafted kidney may be rejected in a patient due to:

- (1) innate immune response.
- (2) humoral immune response.
- (3) cell-mediated immune response.
- (4) passive immune response.

196. Which of the following is/are vector borne disease characterised by swelling of lower limbs and genital region of the body?

- (1) Chikungunya
- (2) Malaria
- (3) Filariasis
- (4) Dengue

197. Which of the following is the primary lymphoid organs of our body?

- (1) Bone marrow
- (2) Pancreas
- (3) Peyer's patches of small intestine
- (4) MALT

198. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Preformed antibodies need to be injected to treat the bite by a viper snake.

Reason R: Antibodies are lipid molecules, each of which has four light chains.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

199. A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is:

- (1) Germ cell internal fallopian transfer.
- (2) Gamete inseminated fallopian transfer.
- (3) Gamete intra fallopian transfer.
- (4) Gamete internal fertilisation and transfer.

200. Read the following statements (A-E).

- A. Yoga has been practised to achieve physical and mental health.
- B. Infectious diseases are very common and everyone suffers from these at some times.
- C. AIDS is an infectious disease.
- D. Healthy persons bring economic prosperity.
- E. Ig D antibody responsible for allergy in human body.

■ ■ ■

How many of the above statements is/are **correct**?

- (1) Two
- (2) One
- (3) Four
- (4) Three



PW Web/App - <https://smart.link/7wwosivoicgd4>

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